

Calculate similarity

$$\text{similarity} = \frac{\text{vector1} \cdot \text{vector2}}{\|\text{vector1}\| \cdot \|\text{vector2}\|}$$

1. **Dot Product:** The numerator of the equation is the dot product (or inner product) of `vector1` and `vector2`:

$$\text{vector1} \cdot \text{vector2} = \sum_{i=1}^n \text{vector1}_i \cdot \text{vector2}_i$$

The dot product is a scalar value that represents the sum of the products of the corresponding elements of the two vectors. It measures the magnitude of the projection of one vector onto the other.

2. **Magnitude (Norm):** The denominator of the equation involves the magnitudes (or norms) of `vector1` and `vector2`. The magnitude of a vector is a measure of its length or size. It is calculated as:

$$\|\text{vector}\| = \sqrt{\sum_{i=1}^n \text{vector}_i^2}$$

The magnitude is the square root of the sum of the squares of the vector elements.

3. **Cosine Similarity:** The equation calculates the cosine similarity between the two vectors. Cosine similarity is a measure of the cosine of the angle between two non-zero vectors. It ranges from -1 to 1, where 1 indicates that the two vectors are pointing in the same direction, -1 indicates that they are pointing in opposite directions, and 0 indicates that they are orthogonal (perpendicular) to each other.

The cosine similarity is calculated as:

$$\text{cosine_similarity} = \cos(\theta) = \frac{\text{vector1} \cdot \text{vector2}}{\|\text{vector1}\| \cdot \|\text{vector2}\|}$$

where θ is the angle between the two vectors.

In the context of text similarity, the cosine similarity is often used to measure the similarity between two text documents represented as vectors. Each document is converted into a vector, where each element represents the frequency or importance of a word in the document. The cosine similarity between these vectors provides a measure of how similar the two documents are in terms of their word content.